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* ZC416 Mathematical Foundations for Machine Learning / Data Science

Webinar-2

AGENDA

(1) Problems

(2) Orthogonal Matrices

(3) Gram-Schmidt orthogonalization

(4) Characteristic polynomial

(5) Eigen values and Eigen vectors

(6) Spectral theorem Singular Value Decomposition

ORTHOGONAL MATRICES

A square matrix "A" is said to be **orthogonal** if its columns are orthonormal.

(**Orthonormal columns** means dot product of any two columns is zero and each column is of length 1.)

Equivalently, we can say

$$AA^T = A^T A = I$$
$$A^{-1} = A^T$$

Remark: Rows of the orthogonal matrices are also orthonormal.

Example:

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

is an orthogonal matrix for all value of θ.

GRAM SCHMIDT PROCESS

- We perform Gaussian elimination on the matrix $\boldsymbol{A}^T\boldsymbol{A}$ where $\boldsymbol{A}$ contains the basis vectors as its columns.
- Upon Gaussian elimination on the augmented matrix we reduce $[\boldsymbol{A}^T\boldsymbol{A}|\boldsymbol{A}^T]$ to get $[\boldsymbol{U}|\boldsymbol{L}^{-1}\boldsymbol{A}^T]$ where $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$.
- Now we shall show that $\boldsymbol{Q}^T = \boldsymbol{L}^{-1}\boldsymbol{A}^T$ is an orthogonal matrix whose rows are orthogonal.
- Consider $\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{L}^{-1}\boldsymbol{A}^T\boldsymbol{A}(\boldsymbol{L}^{-1})^T = \boldsymbol{U}(\boldsymbol{L}^{-1})^T =$ some upper triangular matrix
- But $\boldsymbol{Q}^T\boldsymbol{Q}$ is a symmetric matrix and can only be upper triangular if it is diagonal. Therefore $\boldsymbol{Q}$ is an orthogonal matrix whose columns are orthogonal. They can be normalized to obtain an orthonormal basis.

EXPAMPLE

- Let us start with an example: Consider $\mathbb{R}^2$ and two basis vectors
$$\mathbf{v}_1 = (3, 1)^T \quad \text{and} \quad \mathbf{v}_2 = (2, 2)^T.$$
Put these vectors into columns of a matrix A such that
$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}.$$

- The next step is to perform Gaussian elimination on the following augmented matrix:

$$\left[A^T A \mid A^T\right] = \left[\begin{array}{cc|cc} 10 & 8 & 3 & 1 \\ 8 & 8 & 2 & 2 \end{array}\right].$$

- On performing Gaussian elimination of this augmented matrix we end up with

$$\left[\begin{array}{cc|cc} 1 & 0.8 & 0.3 & 0.1 \\ 0 & 1 & -0.25 & 0.75 \end{array}\right].$$

Note that after the completion of Gaussian elimination the two rows on the right hand side are orthogonal. They form a basis for R2. We can normalize the vectors to get an orthonormal basis.

EXAMPLE

Convert the given basis vectors into orthonormal basis of $\mathbb{R}^3$. Given basis vectors are

$$a = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad c = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Solution:

Let A be a 3×3 matrix containing a, b and c as its columns. Then

$$A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad A^T A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

Now,

$$[A^T A | A^T] = \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 1 & 2 & 1 & 1 & 0 & 1 \\ -1 & 1 & 2 & -1 & 1 & 0 \end{array}\right]$$

Now, apply Gaussian elimination on $[A^T A \mid A^T]$, we have

$$\left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 1 & 2 & 1 & 1 & 0 & 1 \\ -1 & -1 & 2 & -1 & 1 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 + \frac{R_1}{2}]{R_2 \to R_2 - \frac{R_1}{2}} \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 0 & \frac{3}{2} & \frac{3}{2} & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{3}{2} & -1 & \frac{1}{2} & \frac{1}{2} \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + \frac{R_2}{3}} \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 0 & \frac{3}{2} & \frac{3}{2} & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{4}{3} & -\frac{2}{3} & \frac{2}{3} & \frac{2}{3} \end{array}\right]$$

Note that the rows on the right hand side of the above matrix are **mutually orthogonal** and the squares of the lengths of these rows appear on the diagonal of the left side of the matrix. They form a basis for $\mathbb{R}^3$. We can normalize the vectors to get an **orthonormal basis**.

CHARACTERISTIC POLYNOMIAL:

For $\lambda \in \mathbb{R}$ and matrix

$$A \in \mathbb{R}^{n \times n},$$

we can define a polynomial

$$\rho_A(\lambda) = \det(A - \lambda I).$$

This polynomial can be written as

$$\rho_A(\lambda) = c_0 + c_1\lambda + \cdots + c_{n-1}\lambda^{n-1} + (-1)^n c_n \lambda^n$$

where

$$c_0, c_1, \ldots, c_n \in \mathbb{R}.$$

Example: Let the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The characteristic polynomial is

$$\det(A - \lambda I) = (1 - \lambda)^2 - 1.$$

Let $A \in \mathbb{R}^{n\times n}$ be an $n \times n$ square matrix. $\lambda \in \mathbb{R}$ is called an eigenvalue of A and

$$x \in \mathbb{R}^n \setminus \{0\}$$

is the corresponding eigenvector of λ if

$$Ax = \lambda x.$$

This equation is called the eigenvalue equation.

SPECTRAL THEOREM

If A is n x n symmetric matrix then there exists an orthonormal basis of the corresponding vector space V consisting of the eigenvectors of A, and each eigenvalue is real.

SINGULAR VALUES:

The square root of eigenvalues of a symmetric matrix $A^T A$ or AA^T are called singular values of the matrix A.

SINGULAR VECTORS:

The eigen vectors of $A^T A$ corresponding to singular values of A are called singular vectors.

The singular vectors of $A^T A$ are called right singular vectors and singular vectors of AA^T are called left singular vectors.

DEFINITION OF SVD:

Let A be any matrix of order $m \times n$, then this matrix can be decomposed into the product of three matrices given by

$$A_{m\times n} = U_{m\times m}\, \Sigma_{m\times n}\, V^T_{n\times n}$$

and it is called singular value decomposition of A.

Where U and V are called orthogonal matrices or unitary matrices and Σ is called the singular matrix.

SINGULAR VALUE DECOMPOSITION

WORKING RULE:

Step-1: Compute $A^T A$ or AA^T.

Step-2: Find the eigenvalues of $A^T A$ or AA^T.

Step-3: Compute the square roots of the non-zero eigenvalues obtained in the above step. These are called the singular values of A.

Step-4: Construct a singular matrix Σ of the same order as A, having singular values on the diagonal in decreasing order.

Step-5: Find the eigenvectors of A^TA or AA^T and construct an orthogonal matrix U or V consisting of orthonormal eigenvectors of A^TA.

Step-6: Construct the vectors $\sigma_i u_i = Av_i$ and hence obtain the orthogonal matrix U.

Step-7: Finally, $A = U\Sigma V^T$ gives the singular value decomposition (SVD) of matrix A.

Problem

Find Singular value decomposition of the matrix $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$.

Solution: $A_{3\times2} = U_{3\times3}\Sigma_{3\times2}V^T_{2\times2}$
Step-1: Compute A^TA

$$\Rightarrow A^TA = \begin{bmatrix} -3 & 6 & 6 \\ 1 & -2 & -2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}$$

Step-2: Find the eigenvalues of $A^T A$ The characteristic equation (CE) of matrix $A^T A$ is

$$\lambda^2 - \text{Trace}(A^T A)\lambda + |A^T A| = 0 \Rightarrow CE: \quad \lambda^2 - 90\lambda + 0 = 0$$

$$\Rightarrow \lambda(\lambda - 90) = 0$$

$\Rightarrow \lambda_1 = 90, \quad \lambda_2 = 0.$ These are called eigenvalues of matrix $A^T A$.

Step-3: Next find the singular values of A: $\sigma_1 = \sqrt{\lambda_1} = \sqrt{90}$,
$\sigma_2 = 0$

Step-4: Construct the singular matrix

Construct the singular matrix of size same as A and with its diagonal elements being singular values in decreasing order and all other entries are zeros (equivalent diagonal matrix).

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{90} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Step-5: Find the eigenvectors of A^TA. The eigenvectors of A^TA are called right singular vectors of A with respect to each eigenvalue. For $\lambda = 90$

$$\Rightarrow A^TA - 90I = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix} - 90 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -9 & -27 \\ -27 & -81 \end{bmatrix}$$

Hence from first row we get

$$-9x - 27y = 0$$

$\Rightarrow x = -3y = -3k; y = k$ is a free variable.
Therefore eigenvectors with respect to $\lambda = 90$ is

$$X_1 = k \begin{bmatrix} -3 \\ 1 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

For $\lambda = 0$

$$\Rightarrow A^TA - 0I = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix} - 0\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}$$

$$\Rightarrow y = k \text{ is a free variable.}$$

Hence from first row we get

$$81x - 27y = 0$$

$$\Rightarrow x = \frac{1}{3}y = \frac{1}{3}k$$

Therefore eigenvectors w.r.t. $\lambda = 0$ is $X_2 = \frac{k}{3}\begin{bmatrix} 1 \\ 3 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

Step-6: Write the orthogonal matrix.

Write the orthogonal matrix or right singular vectors matrix

$$V = [\hat{v}_1 \ \hat{v}_2]$$

$$V = \begin{bmatrix} \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}$$

Step-7: Find the left eigenvectors or left singular vectors

Find the left eigenvectors or left singular vectors $\hat{u}_i$ using

$$\hat{u}_i = \frac{Av_i}{\sigma_i}$$

For $i = 1$

$$u_1 = \frac{Av_i}{\sigma_i} = \frac{1}{\sqrt{90}} \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} \frac{3}{\sqrt{10}} \\ \frac{-1}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} \frac{-10}{\sqrt{900}} \\ \frac{20}{\sqrt{900}} \\ \frac{20}{\sqrt{900}} \end{bmatrix}$$

Since the second eigenvalue is zero,

$$\lambda_2 = \sigma_2 = 0$$

to find the other two vectors let us construct the orthogonal vectors using orthogonality conditions.

Let the next vector be $w = (x, y, z)$ then $w \cdot u_1 = 0$

$$\Rightarrow 10x - 20y - 20z = 0,$$

$\Rightarrow y = k_1 \quad$ and $\quad z = k_2$
are free variables, hence

$$x = 2y + 2z = 2k_1 + 2k_2, \Rightarrow w = k_1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Hence the vectors orthogonal to u_1 are $w_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $w_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

Let $u_2 = w_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \hat{u}_2 = \frac{u_2}{\|u_2\|} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}$

$$u_3 = w_2 - \frac{\langle w_2, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} - \frac{4}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 2 \\ -4 \\ 5 \end{bmatrix}$$

$$\Rightarrow u_3 = \begin{bmatrix} 2 \\ -4 \\ 5 \end{bmatrix} \Rightarrow \hat{u}_3 = \frac{u_3}{\|u_3\|} = \begin{bmatrix} \frac{2}{\sqrt{45}} \\ -\frac{4}{\sqrt{45}} \\ \frac{5}{\sqrt{45}} \end{bmatrix}$$

Hence the orthogonal matrix U or matrix of left singular vectors is given by

$$U = \begin{bmatrix} \frac{10}{\sqrt{900}} & \frac{2}{\sqrt{5}} & \frac{2}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & \frac{1}{\sqrt{5}} & -\frac{4}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & 0 & \frac{5}{\sqrt{45}} \end{bmatrix}$$

Step-8:

Hence the singular value decomposition of given matrix A is

$$A = U\Sigma V^T$$

$$= \begin{bmatrix} \frac{10}{\sqrt{900}} & \frac{2}{\sqrt{5}} & \frac{2}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & \frac{1}{\sqrt{5}} & -\frac{4}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & 0 & \frac{5}{\sqrt{45}} \end{bmatrix} \begin{bmatrix} \sqrt{90} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}$$

PROBLEM

(A) Given the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 5 & 2 \end{bmatrix},$$

a professor asks two of his best students enrolled in Linear Algebra class to find the maximum value of $\mathbf{x}^T\mathbf{A}\mathbf{x}$, subject to the fact that $\|\mathbf{x}\|^2 = 1$, where $\|\|$ is the Euclidean norm. Given that the students have not studied Calculus earlier, the first student says that this is impossible, whereas the second one is optimistic in estimating the value. Who is correct and why? Give adequate justifications.

HINT: Find a symmetric matrix $\mathbf{B}$ such that $\mathbf{x}^T\mathbf{A}\mathbf{x} = \mathbf{x}^T\mathbf{B}\mathbf{x}$

(B) Is $\lambda = 4$ an eigenvalue of

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}?$$

If yes, find the corresponding eigenvector.

Answers

(A) expanding

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

and rearranging terms gives us the off diagonal terms to be $(5 + 3)/2 = 4$ for the symmetric matrix. Thus,

$$\mathbf{B} = \begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix}$$

Let λ_1, λ_2 be eigenvalues of $\mathbf{B}$ with $\mathbf{e}_1, \mathbf{e}_2$ being the corresponding eigenvectors. Any $\mathbf{x}$ in $\mathbb{R}^2$ can be written as:

$$\mathbf{x} = p * \mathbf{e}_1 + q * \mathbf{e}_2,\ p^2 + q^2 = 1$$

Now,

$$\mathbf{x}^T\mathbf{B}\mathbf{x} = \begin{bmatrix} p\mathbf{e}_1 & q\mathbf{e}_2 \end{bmatrix} \begin{bmatrix} p\mathbf{e}_1\lambda_1 \\ \mathbf{e}_2\lambda_2 \end{bmatrix} = p^2\lambda_1 + q^2\lambda_2 \leq \max(\lambda_1, \lambda_2)$$

$$\lambda_{max} = \frac{3+\sqrt{65}}{2}$$

(B) If $\lambda = 4$ is an eigenvalue, then the eigenvector is in the null space of $\mathbf{A} - \lambda \mathbf{I}_3$. For this, the null space should have dimension > 0 which can be verified as:

$$\left[\begin{array}{ccc|c} 3-4 & 0 & -1 & 0 \\ 2 & 3-4 & 1 & 0 \\ -3 & 4 & 5-4 & 0 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 * -1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 2 & -1 & 1 & 0 \\ -3 & 4 & -1 & 0 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2 * R_1 , R_3 \leftarrow R_3 + 3 * R_1}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 4 & 4 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Corresponding eigenvector is: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} t$

PROBLEM

Given the characteristic equation of a matrix A, can we compute the characteristic equation of cA where c is a non-zero scalar without knowing the entries of A? If so, show how to do it using detailed calculations. Otherwise explain why it is not possible. Clearly state all your assumptions

SOLUTION

Let $\mu = \frac{\lambda}{c}$, and assume that the characteristic equation

$$\det(A - \mu I) = 0$$

can be written in terms of the polynomial

$$\mu^n + a_{n-1}\mu^{n-1} + \cdots + a_1\mu + a_0 = 0.$$

Substituting $\mu = \frac{\lambda}{c}$ in this polynomial, we get

$$\left(\frac{\lambda}{c}\right)^n + a_{n-1}\left(\frac{\lambda}{c}\right)^{n-1} + \cdots + a_1\left(\frac{\lambda}{c}\right) + a_0 = 0.$$

Multiplying throughout by c^n, we finally get

$$\lambda^n + ca_{n-1}\lambda^{n-1} + \cdots + a_1c^{n-1}\lambda + a_0c^n = 0.$$

Thus, we see that the characteristic equation of cA can be obtained by taking the coefficient a_k of the kth term of the characteristic equation of A and multiplying it by c^{n-k}. Therefore, there is no need to look at the entries of the matrix A.

PROBLEM

Given that the matrix

$$A = \begin{bmatrix} 1 & 0 & a \\ 0 & 2 & 0 \\ a & 0 & b \end{bmatrix}, \quad a, b \in \mathbb{R}$$

has 2 and -1 as its eigenvalues with algebraic multiplicity of 2 and 1 respectively. Further, the eigenvector corresponding to eigenvalue -1 is

$$\mathbf{e}_3 = \begin{bmatrix} 1 \\ 0 \\ -\sqrt{2} \end{bmatrix}$$

(a) Find the value of b.

(b) Write all the eigenvalues and their geometric multiplicity.

(c) By observation, and using the properties of a symmetric matrix, find the other two eigenvectors ($\mathbf{e}_1$ and $\mathbf{e}_2$) of A.

(d) Write the spectral decomposition of the matrix A.

(e) Find the value of a.

SOLUTION

(a) The eigenvalues of A are $2, 2, -1$. Hence,

$$1 + 2 + b = 2 + 2 - 1.$$

Thus, $b = 0$.

(b) Since A is symmetric, eigenvalue 2 has GM 2 and eigenvalue -1 has GM 1.

(c) The eigenvector corresponding to eigenvalue -1 is given by

$$\mathbf{e}_3 = \begin{bmatrix} 1 \\ 0 \\ -\sqrt{2} \end{bmatrix}.$$

Solution

The other two eigenvectors correspond to eigenvalue 2 and both should lie in a plane orthogonal to $\mathbf{e}_3$. Thus, $\mathbf{e}_1$ and $\mathbf{e}_2$ are:

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \sqrt{2} \\ 0 \\ 1 \end{bmatrix}$$

(d) The spectral decomposition of A is:

$$A = \begin{bmatrix} 0 & \frac{\sqrt{2}}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & -\frac{\sqrt{2}}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ \frac{\sqrt{2}}{\sqrt{3}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{\sqrt{2}}{\sqrt{3}} \end{bmatrix}.$$

SOLUTION

Solution

(e) As e_3 is an eigenvector corresponding to $\lambda = -1$, so we have

$$A.e_3 = -1.e_3$$

$$\Rightarrow \left(\begin{bmatrix} 1 & 0 & a \\ 0 & 2 & 0 \\ a & 0 & b \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) .e_3 = 0$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & a \\ 0 & 3 & 0 \\ a & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 \\ 0 \\ -\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a = \sqrt{2}.$$

Problem

Given that the Singular Value Decomposition of A is

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & e_{22} \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & \lambda_2 & \gamma_2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{3\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2\sqrt{2}}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

where

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

(a) Find the value of e_{22}.

(b) Find the value of λ_2.

SOLUTION

(a)

$$e_{22} = -\frac{1}{\sqrt{2}}.$$

(b) Utilizing the given decomposition of matrix A and comparing the corresponding elements on both sides we get ,

$$\frac{1}{\sqrt{2}}.\frac{5}{\sqrt{2}} + \frac{\lambda_2}{6} = 3$$

this gives, $\lambda_2 = 3$

SOLUTION

PROBLEM

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(a) Obtain the left-singular vectors of A.

(b) Obtain the right-singular vectors of A.

(c) Obtain the singular value matrix Σ. What is the spectral norm of A?

SOLUTION

(a)

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

SOLUTION

Eigenvalues and eigenvectors of AA^T:

$$\lambda_1 = \sigma_1^2 = 6, \qquad u_1' = \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}, \qquad \|u_1'\| = \sqrt{30}$$

$$\lambda_2 = \sigma_2^2 = 1, \qquad u_2' = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix}, \qquad \|u_2'\| = \frac{\sqrt{5}}{2}$$

$$\lambda_3 = \sigma_3^2 = 0, \qquad u_3' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \qquad \|u_3'\| = \sqrt{6}$$

SOLUTION

Singular value matrix:

$$\Sigma = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & \sqrt{1} \\ 0 & 0 \end{bmatrix}$$

Left singular vectors:

$$U = \left[u_1 = \frac{u_1'}{||u_1'||} \quad u_2 = \frac{u_2'}{||u_2'||} \quad u_3 = \frac{u_3'}{||u_3'||} \right]$$

SOLUTION

Right Singular Vectors:

$$v_1 = \frac{1}{\sigma_1} A^T u_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} \frac{1}{\sqrt{30}} = \frac{1}{\sqrt{6}\sqrt{30}} \begin{bmatrix} 12 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$v_2 = \frac{1}{\sigma_2} A^T u_2 = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix} \frac{2}{\sqrt{5}} = \begin{bmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

SOLUTION

Thank you